

Gamma detection for source localisation in targeted alpha therapy

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Abstract. Targeted alpha therapies can deliver a highly localised dose to tumours, but treatment planning and effectiveness depend on whether the radionuclide source remains near the intended site after administration. In the AlphaGlue delivery concept, alpha emitters are applied to a tumour region in a thin glue layer. We present a Geant4 simulation study showing that decay-chain gamma detection can distinguish retained, broadly diffused and locally displaced source states in an AlphaGlue-like geometry. 50 repeated simulations of 10,000 primary events were generated, and baseline-subtracted gamma-hit profiles were successfully used to reconstruct the broad diffusion fraction with $R^2 = 0.9987$ and a mean absolute error of about one percentage point. Displaced source activity was detected by gamma signal analysis and source position was reconstructed with mean position errors of 16–30 mm. These results support gamma detection as a candidate method for emitter-position monitoring, treatment planning and delivery verification.

1 Introduction

In targeted alpha therapy, the delivered dose is strongly local. A treatment plan therefore depends on the initial activity distribution and its evolution after administration. The AlphaGlue concept proposes suspending alpha emitters in a thin glue layer applied to tissue [1]. The concept has previously been evaluated with Geant4 and Geant4-DNA simulations for dose and DNA-damage modelling [2–7]. Its delivered dose depends on whether source material and progeny remain near the intended deposition site, diffuse into surrounding tissue or are redistributed by biological transport [8, 9]. This study investigates whether the accompanying gamma signal can be used as a theranostic observable to monitor emitter position, detect source movement and provide useful information as a function of detection time.

2 Simulation model

Photon transport was simulated with the Geant4 toolkit [2]. The model places a 40 mm × 2 mm × 40 mm loaded glue layer on the +y surface of a 20 cm × 20 cm × 150 cm water phantom. Six detector panels surround the phantom at 20 cm from the water surface. Each photon entry into a detector panel is counted as one hit.

Radionuclides or their progeny may move away from the applicator via diffusion or perfusion due to blood circulation. This migration can produce a broad redistribution through the surrounding tissue, with large variation reported between experiments [8, 9]. To test whether these behaviours can be distinguished from source retention, seven source configurations were simulated. The retained-source configuration was used as the baseline. Broad redistribution was represented by cases in which 10%, 50% or 90% of the source was distributed throughout the

water volume. Localised transport was represented by placing 20%, 40% or 60% of the source in a 5 cm-side water cube centered at $(x, y, z) = (0, 0, 500 \text{ mm})$, spanning $475 \leq z \leq 525 \text{ mm}$.

The geometry is described in Fig. 1, along with example detector displays showing the three clinical scenarios. Each configuration was simulated with 50 independent repetitions of 10,000 primary decays, giving 500,000 simulated decays per configuration and 3.5 million simulated decays across all source states. The repeated runs define the run-to-run uncertainty used in the residual and detectability analysis. The simulated decays are rescaled to a DaRT-like reference activity of $3 \mu\text{Ci}$ (111 kBq), matching the order of a single DaRT seed [10].

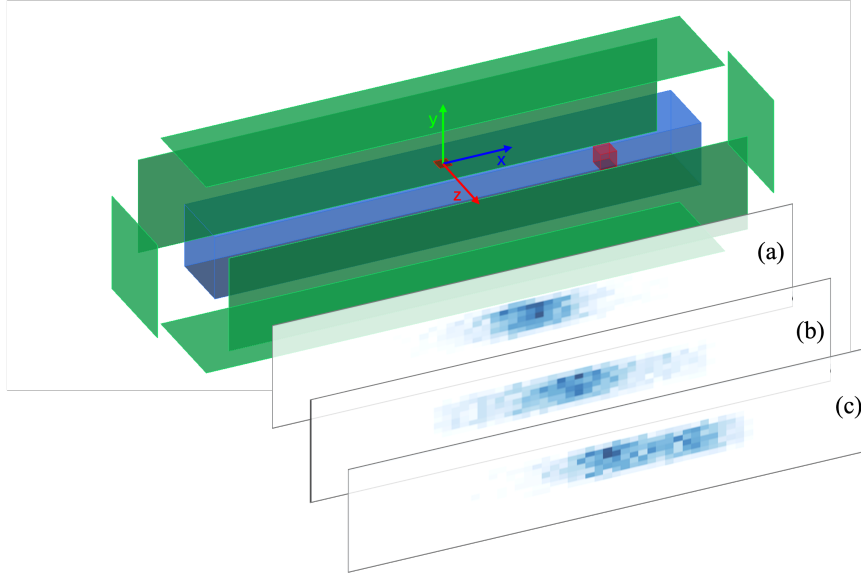


Figure 1: Simulation geometry showing the water phantom (blue), AlphaGlue applicator region (red, centre), surrounding detector panels (green), a displaced source (red, right side) and three offset example x -panel projections: baseline (a), broad redistribution (b), and displaced localisation (c).

3 Baseline subtraction

Each photon entry into a detector panel was recorded as one hit, together with its detector panel and entry position. Detector panels are named according to the side of the phantom on which they are located. The $+x$ and $-x$ panels are the two lateral panels at positive and negative z ; the $+y$ panel faces the AlphaGlue applicator and the $-y$ panel lies on the opposite side. The $+z$ and $-z$ panels close the longitudinal ends of the geometry. A combined x -panel profile is obtained by summing the $+x$ - and $-x$ -panel counts bin by bin and plotting them against the longitudinal coordinate z . Thus, “ x -panel” identifies the detector location, while z identifies position along the panel. Longitudinal profiles were examined because redistribution changes the spatial pattern of detector hits more strongly than the total number of hits. The total detector-entry rate differed by only 3.8% between the retained baseline and the 90% broad-diffusion case, whereas subtraction of the retained-source profile revealed clear spatial differences (Fig. 2). The retained source defines the reference profile $\bar{h}_p^B(z)$ for detector panel p . Candidate source states are represented by

$$\Delta h_p(z) = h_p(z) - \bar{h}_p^B(z), \quad (1)$$

with the repeated baseline simulations defining the bin-wise noise scale. Each profile was divided by the number of simulated primary decays and rescaled to the initial ^{222}Rn population corresponding to the $3\ \mu\text{Ci}$ reference activity. The plotted residual is the difference between the mean activity-scaled candidate and retained-source profiles. Positive residuals mark detector regions where the candidate produces more gamma entries than the retained-source case, while negative residuals mark depleted regions. In the combined (x)-panel projection, broad diffusion widens the residual pattern, whereas localised displacement produces a shifted excess near the displaced source region. Changes in the balance between the glue-facing and opposite detector panels provide an additional feature for the quantitative analysis.

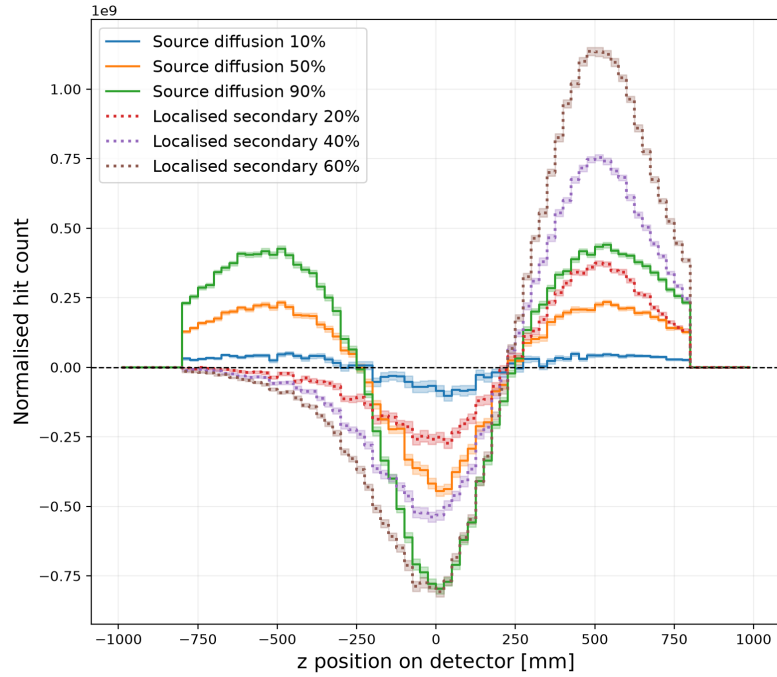


Figure 2: Activity-scaled, baseline-subtracted combined $+x$ and $-x$ detector profiles for broad diffusion and localised displacement. Shaded bands show variation across repetitions.

As an illustration of the general ROI strategy, Fig. 3 compares the full longitudinal profile formed by summing the $+x$ and $-x$ panel counts with a region at $-750 \leq z \leq -550$ mm, where little signal is expected from retained activity. An excess in this region indicates that the detected distribution has spread beyond the applicator footprint. This ROI illustrates the principle on which the quantitative estimators in Section 4 are based: source redistribution is inferred from spatial changes relative to the retained-source reference.

4 Signal analysis and source redistribution metrics

The signal analysis addresses four tasks: detecting and locating a compact secondary source, estimating the broadly diffused fraction when no compact source is present, estimating how much signal remains consistent with the applicator region, and determining the observation time required to distinguish each candidate from the retained-source baseline. Together, these tasks determine whether activity has moved, whether the redistribution is broad or localised, where a compact displaced source is located, and how quickly each change becomes detectable.

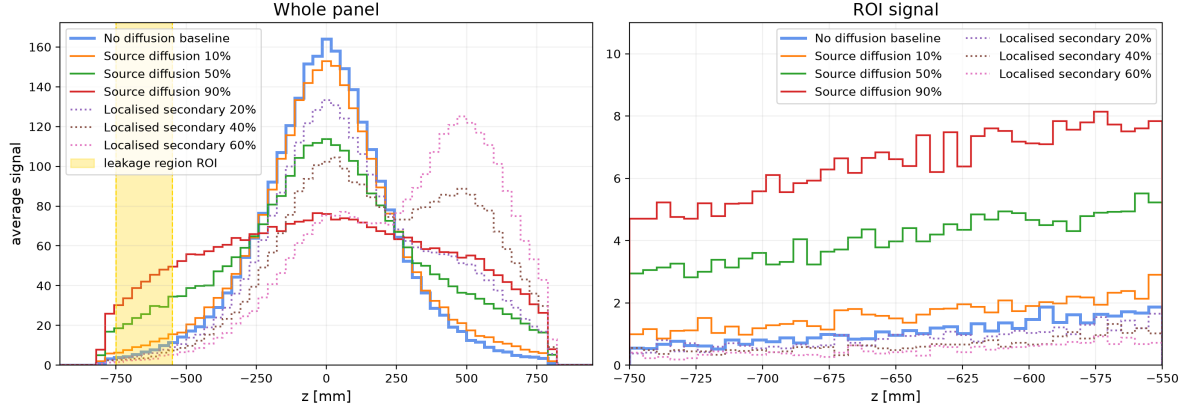


Figure 3: Combined $+x/-x$ longitudinal profiles (left) and an illustrative low-signal ROI (right). The ROI highlights the broad-diffusion excess outside the retained-source region.

Both broad diffusion and compact displacement can reduce the central signal, so the compact-secondary test is applied first. A negative result enables the broad-diffusion and applicator-retention estimates; a positive result instead triggers source localisation.

4.1 Compact secondary-source detection and localisation

Counts and spatial-profile features from the four side panels were used to distinguish a compact displaced source from retained and broadly redistributed source states. The panel asymmetries were calculated as

$$A_i = \frac{N_{+i} - N_{-i}}{N_{+i} + N_{-i}}, \quad i \in \{x, y\},$$

where N_{+i} and N_{-i} are the total numbers of entries on the opposing side panels. Broad diffusion primarily changes A_y , because activity moves away from the applicator-facing $+y$ panel, while a source displaced along z produces a shifted feature within the longitudinal detector profiles. Compact-source detection used a balanced logistic-regression classifier supplied with standardised panel asymmetries, baseline-subtracted profile bins, positive-residual integrals, peak and centroid positions, longitudinal widths, ROI rates and the total detected rate. Performance was evaluated using leave-one-repetition-out validation.

A displaced-source ROI of $250 \leq z \leq 750$ mm was defined in the combined $+x/-x$ and $+y/-y$ longitudinal profiles. For each view, the ROI signal was obtained by summing the positive candidate-minus-baseline residual over all z bins in this interval. After positive detection, the source position was estimated from the positive-residual centroid separately in the combined x - and y -panel profiles. The final position estimate was the mean of these two centroids.

$$\hat{z}_{\text{sec}} = \frac{\hat{z}_{X\text{sum}} + \hat{z}_{Y\text{sum}}}{2},$$

The reconstructed positions and their repetition-level residuals are reported in Fig. 6.

4.2 Broad-diffusion fraction

Broad diffusion is estimated only after the compact-secondary test is negative. Inputs comprise shape-normalised z -profile bins from the combined x , combined y and all-panel views; the standard deviation and central-80% interquartile width ($z_{90} - z_{10}$); the central $+y$ fraction; x - and y -panel asymmetries; and positive-residual integral and centroid summaries. These features are

standardised and supplied to ridge regression. The ridge-regression regularisation strength was selected by internal cross-validation over $\alpha = 10^{-4}$ to 10^4 , balancing agreement with the training data against model complexity. The model is trained only on the retained and 10%, 50% and 90% broad-diffusion states; localised-source cases are excluded. Performance is evaluated with leave-one-repetition-out validation. The resulting predictions are shown in Fig. 5.

4.3 Applicator-retained signal

The applicator-facing observable is the $+y$ -panel count in the central ROI, $|z| \leq 150$ mm, normalised per primary event. Dividing this rate by the retained-source rate gives a baseline-relative proxy for the fraction of signal remaining at the applicator. The all-panel central ROI is retained as a cross-check. As shown in Fig. 4, both broad diffusion and compact displacement reduce this signal, so an applicator-retained fraction is quoted only after the compact-secondary test has been evaluated.

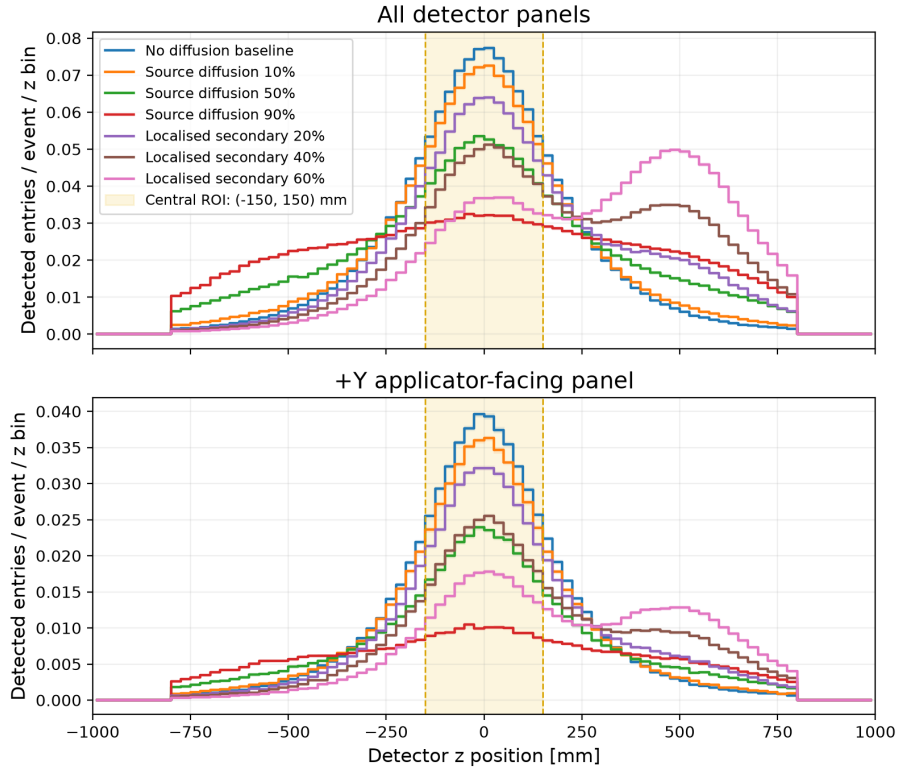


Figure 4: Central $|z| \leq 150$ mm ROI for all detector panels (top) and the applicator-facing $+y$ panel (bottom). The $+y$ ROI rate provides the baseline-relative applicator-retained signal.

4.4 Time-window detectability

The same baseline-subtracted profiles and task-specific ROIs are accumulated in increasing observation windows. At each window, the candidate signal is compared with the distribution obtained from the baseline repetitions. A source state is considered detectable when the candidate–baseline separation exceeds three standard deviations of the run-to-run baseline variation. The reported detection time is the first cumulative window that crosses this threshold.

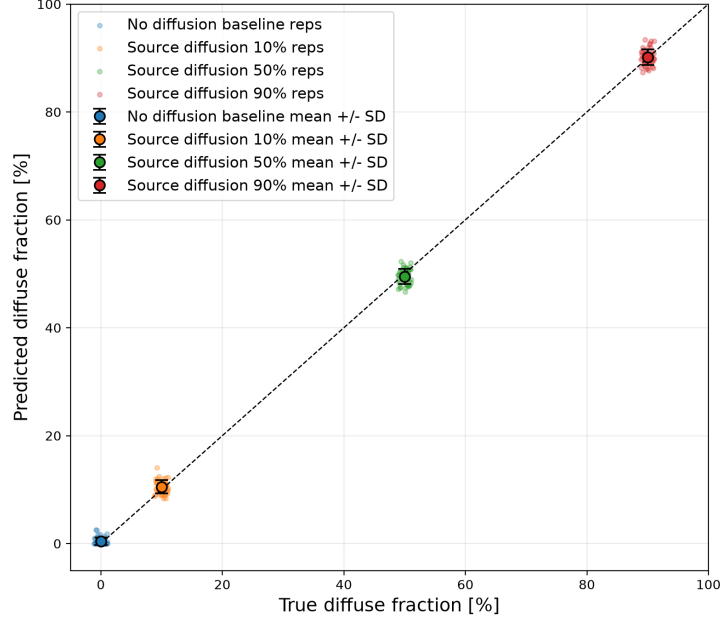


Figure 5: Diffusion fraction inferred from 50 independent repetitions. Small points show individual repetitions and large markers show the mean and standard deviation.

This procedure tests when the spatial redistribution used by the diffusion and localisation estimators becomes statistically resolvable; the resulting threshold times and an example cumulative profile are presented in Section 5.2.

5 Results

5.1 Signal inference

Using profile-shape features from repeated simulations, the broad diffusion fraction was recovered with $R^2 = 0.9987$ and a mean absolute error of about one percentage point (Fig. 5). The baseline-normalised central $+y$ -panel ROI signal decreased monotonically with broad diffusion, to 0.925 ± 0.017 , 0.618 ± 0.013 and 0.289 ± 0.009 for the 10%, 50% and 90% broad-diffusion cases, respectively (Fig. 4). The ratio measures the central-ROI detector signal relative to the retained-source baseline; it is not a direct measurement of the fraction of activity remaining in the applicator. The compact-source strategy identified the localised displaced source in all 20%, 40% and 60% localised-displacement cases. The reconstructed displaced-source positions were approximately $z = 516$ mm, 528 mm and 530 mm, respectively, compared with the simulated source centre at $z = 500$ mm. Across these localised fractions, mean position errors were 16–30 mm, with the repetition-level residual distributions shown in Fig. 6. The repetition-to-repetition spread decreased from 6.6 mm to 2.1 mm as the displaced-source fraction increased, while the mean estimate remained shifted towards positive z . This indicates increasing precision but a systematic bias in the positive-residual centroid estimator; calibration using simulations at multiple source positions will be required in future work.

5.2 Time dependence

Cumulative time-window profiles show that the baseline-subtracted signal is already visible on hour-scale windows and becomes stronger over longer observation windows. Across the 50

simulation repetitions, detectability was defined as candidate-baseline separation greater than 3σ of the run-to-run baseline variation. The 50–90% broad-diffusion and 20–60% localised-displacement cases crossed threshold in about 1–2 hours, while the weaker 10% broad-diffusion case required about 6 hours. A representative time-window profile is shown in Fig. 7. A $t < 10^6$ s analysis cut suppresses a late decay-chain tail while retaining the early signal relevant to the simulated source-localisation window. The first threshold-crossing times are summarised in Table 1.

Table 1: First 3σ detection times from the time-window analysis of 50 independent repetitions.

Case group	First 3σ detection
10% broad diffusion	about 6 h
50–90% broad diffusion	about 1–2 h
20–60% localised displacement	about 1–2 h

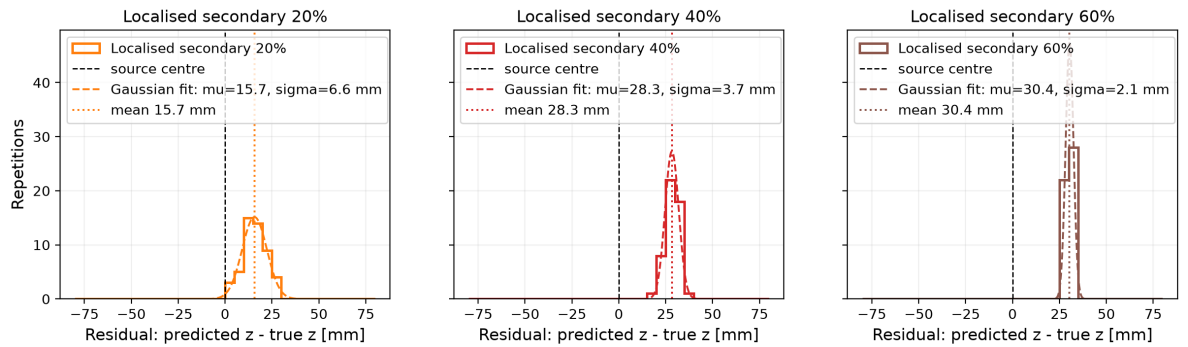


Figure 6: Localisation residuals across 50 repetitions for each displaced-source fraction. The dashed black line marks the true source centre; coloured lines indicate the mean and Gaussian fit.

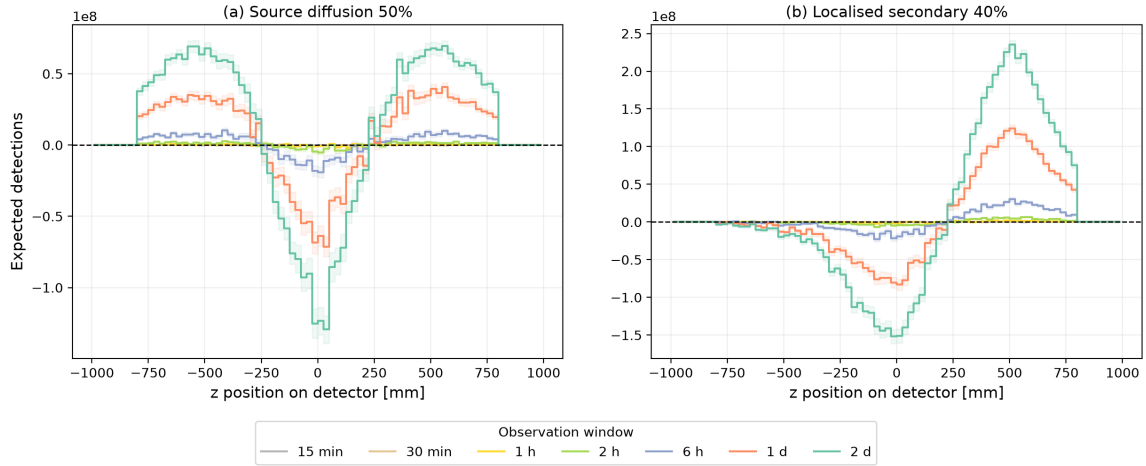


Figure 7: Cumulative combined $+x/-x$ residual profiles for 50% broad diffusion (left) and 40% localised displacement (right). Broad diffusion produces a symmetric redistribution away from the centre, whereas localised displacement produces an asymmetric excess near the displaced-source region. These signatures first exceed the 3σ detection threshold after approximately 2 hours and 1 hour, respectively.

6 Conclusion

Decay-chain gamma detection provides source-location information and broad redistribution estimation. Baseline-subtracted gamma profiles distinguished retained, broadly redistributed and locally displaced source states. Broad-diffusion fraction was reconstructed with $R^2 = 0.9987$ and a mean absolute error of approximately one percentage point. All simulated compact-source cases were detected, with mean reconstructed positions 16–30 mm from the simulated source centre. Depending on the source state, redistribution became detectable after approximately 1–6 hours. Broad diffusion and compact source displacement were treated as independent source states in the present analysis. A combined case, in which diffuse redistribution and a compact displaced source occur together, should be included in the next simulation and inference studies. Further work will also include detector response, background estimation and patient-scale geometries so that activity-normalised profiles can be interpreted as experimentally measurable count rates.

References

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